



EEE 4861 Assignment

Design and Investigation of a Simple Sensor Circuit Intended for Smart Food Packaging on Biodegradable Packaging

Section B

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Submitted to,
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Motivation: In Bangladesh, many food and daily grocery items lack proper packaging, leading to issues such as post-harvest losses and compromised food safety. Smart food packaging offers a solution by integrating sensor systems that provide early alerts about food quality. This not only enhances food safety and extends shelf life but also improves traceability and transparency, promotes greater consumer engagement, and supports sustainability through reduced waste.

Objective and CEP statement: Individuals or groups of up to three students will design and investigate a simple sensor circuit intended for smart food packaging on biodegradable packaging.

- ✓ The sensor circuit must be implemented using the Electrical and Electronic Engineering knowledge and all the modern tools studied in the class: SPICE simulator, stick diagram, layout design, and performance analysis using SPICE.
- ✓ A key component of the assignment is the submission of a detailed proposal that includes a thorough literature review and the selection of a suitable biodegradable plastic packaging material.
- ✓ The assignment must include a proposal to demonstrate how the designed sensor circuit can be integrated into the chosen biodegradable packaging to monitor food quality over time, thereby addressing food safety and environmental concerns.
- ✓ Additionally, students must propose a structured decision-making algorithm or framework that supports actions prioritizing food safety, human well-being, and sustainable practices under the Bangladesh Environmental Conservation Rules 2023(4).
- ✓ A comprehensive design report must be submitted upon completion of the assignment.

Critical Challenges: A common engineering practice is only the use barcode on the conventional plastic package to calculate product bill during payment. Conventional plastic packaging will not be used as they pollute the environment. Also, use of environmental-friendly electronic circuit system will enhance quality of life on earth.

Conflicting Requirements: Students are required to explore various design methodologies to evaluate potential solutions although such expensive biodegradable packaging with smart sensor system is difficult to integrate in a low-income country. Also, a lack of health safety knowledge is also one of the key constraints to implement the proposed system.

1. Course Outcomes (COs): EEE 4861 (VLSI Design)

CO4: Engage in independent research with regards to emergent topics in VLSI domain.

CO5: Consider safety and health issues in design of food packaging on biodegradable packaging.

2. Program Outcomes (POs): EEE 4861 (VLSI Design)

PO5: (Modern tool usage) Create, select and apply appropriate techniques, resources, and **modern engineering and IT tools**, including prediction and modelling, to complex engineering problems (CEP), with an understanding of the limitations. (K6)

PO7: (Environment and Sustainability) Understand and evaluate the **sustainability** and impact of professional engineering work in the solution of complex engineering problems in societal and **environmental** contexts. (K7).

3. We explore how a few Complex Engineering Problems (Ps) through this assignment:

P1: This assignment requires depth of Electrical and Electronic Engineering knowledge (K3) and also requires study of existing smart food packaging models with similar Goal (K8).

P2: Conflicting technical requirements: Plastic packaging is available, however, biodegradable plastic packaging are expensive with challenges in sensor system integration onto it (K5, K6). A low price variant is critical to achieve.

P3: No obvious or specific sensor circuit. A Ring oscillator, inverter, even voltage divider circuit can be a temperature sensor, some other circuits exhibits performance variation in the presence of O₂, N₂ and even under toxic environment (K4).

P4: Electrical and Electronic Engineering students are not typically study to issues related to bio-safety and food safety.

4. Knowledge Profiles (Ks) through this assignment:

✓ We accumulate **K3~K8**.

5. Complex Engineering Activities (As) through this assignment:

✓ We claim no Complex engineering activity from this assignment.

CO-PO-K-P-A Mapping: for CEP

COs	POs	Knowledge Profile(s)	CEP	CEA
CO4	PO5	K6	P1, P2, P3, P4	-
CO5	PO7	K7	-	-

6. Rubrics for Assessment: (75 marks)

Sl.	Section	Description	Marks
01.	Literature Review & Material Selection (CO4, PO5)	-Thorough literature review on sensor circuits for food packaging. -Selection and justification of suitable biodegradable plastic packaging material.	10
02.	Sensor Circuit Design & Tools Usage (CO4, PO5)	-Application of EEE knowledge in design. -Proper use of SPICE simulator, stick diagram, layout design. - Performance analysis using SPICE.	20
03.	Integration with Packaging (CO5, PO7)	-Proposal on integrating the designed sensor into biodegradable packaging. -Practical considerations for monitoring food quality over time.	10
04.	Decision-Making Algorithm / Framework (CO5, PO7)	-Structured decision-making method for prioritizing food safety, human well-being, and sustainability. -Alignment with Bangladesh Environmental Conservation Rules 2023(4).	10
05.	Proposal Quality and originality (CO4, PO5)	-Clarity, technical depth, and feasibility of the initial proposal. -Logical flow and completeness.	10
06.	Final Design Report (CO4, PO5)	- Comprehensive and well-organized documentation. -Includes all results, design files, simulations, and conclusions. - Clear communication of work. - Creativity, innovation, and originality in approach.	15

Abstract:

This study introduces a low-cost and biodegradable smart food packaging system with the purpose of improving food safety and reducing post-harvest loss. The system incorporates a temperature sensor inside polylactic acid (PLA) packaging that can provide real-time temperature deviation detection that could affect food quality. The core of the sensor employs a relaxation oscillator based on a 6-transistor CMOS Schmitt trigger circuit, which senses temperature fluctuation by varying its oscillation frequency. The frequency is converted into a freshness index that facilitates early detection of spoilage as well as extends the shelf life for perishable goods. PLAs have the benefit of biodegradability as well as adherence to the Environmental Conservation Rules 2023 of Bangladesh, ensuring support for green initiatives. The proposed solution also uses energy-conserving techniques, such as thin-film batteries and energy harvesting, to minimize environmental footprints as well as enhance traceability as well as food safety.

Motivation

Fresh vegetables and fruits of Bangladesh often undergo high spoilage due to the lack of cold-storage facilities, ineffective control during transit, and improper packaging. All these contribute to high microbial growth within a short period of time, leading to wastage of food, loss of income to the grower and potential harm to consumer's health. Passive protection of traditional packages does not provide any time-dependent information about the storage environment and exposes sensitive products to unnoticed temperature fluctuations that degrade quality and safety.

To address these challenges, this project proposes the implementation of a low-cost biodegradable smart packaging system with an onboard temperature sensor employed with a relaxation oscillator with Schmitt trigger. Through real-time monitoring of the temperature swing within the package, the system makes the packaging itself an active element in cold-chain control. The sensor enable real-time monitoring of storage conditions deviations from the best, particularly for coldstorage facilities aimed at preserving freshness. For instance, exposure for long periods above 4 °C can be reported on, enabling timely intervention before the spoilage.

This sensor-enabled packaging fulfills several critical needs it deliver improved food safety through pre-warning of temperature deviations that can induce spoilage it achieve shelf life through facilitation of stakeholders in maintaining proper cold-storage conditions; it assist traceability and transparency along the supply chain by linking sensor readings to storage history; and it facilitate consumer education through simple-to-interpret visual or digital cues.

Its use as the packaging material is eco-friendly. PLA is nontoxic, compostable, and biodegradable food contact material that can be used to accommodate electronic sensors without affecting structural properties. The action reduces the use of ordinary plastics, conserves waste, and is compliant with the Bangladesh Environmental Conservation Rules 2023(4).

With the integration of biodegradable packaging and an integrated low-power temperature sensor, this paper provides a scalable, practical solution for post harvest loss mitigation, for ensuring food quality for cold-storage systems, and for starting environmentally friendly operations in Bangladesh's food chain. The proposed smart packaging system not only address the essential requirements of food safety but also contribute to long-term environmental and operation optimization.

Objectives and Complex Engineering Problems (CEP) Addressed

The primary objective of this project is to design, implement, and investigate a low-cost, biodegradable smart food packaging system capable of monitoring temperature in cold-storage conditions to prevent spoilage of fruits and vegetables. Specifically, the system integrates a relaxation-oscillator-based temperature sensor, implemented with a 6-transistor CMOS non-inverting Schmitt trigger and inverter, into polylactic acid (PLA) packaging. The sensor continuously tracks temperature variations and provides a calibrated freshness index, enabling timely intervention and decision-making to maintain food quality.

Additional objectives include:

- Ensuring practical integration of the sensor circuit with PLA packaging without compromising mechanical strength, biodegradability, or user safety.
- Developing a calibrated mapping between temperature readings and spoilage risk (Safe, Warning, Unsafe), particularly suitable for cold-storage conditions.
- Exploring energy-efficient power solutions (thin-film battery or energy harvesting) to enable continuous or periodic monitoring with minimal cost and environmental impact.
- Demonstrating a prototype that validates the proposed methodology through simulation (SPICE), layout design, and performance analysis.

Addressing Complex Engineering Problems (CEP)

P1 – Depth of Electrical and Electronic Engineering Knowledge:

The project leverages advanced EE knowledge in analog and digital circuit design, including relaxation oscillators, Schmitt triggers, RC networks, and frequency-to-digital conversion. It also incorporates study of existing smart packaging systems and sensor designs to inform circuit choice and calibration, ensuring that the solution is grounded in both theoretical and practical understanding.

P2 – Conflicting Technical Requirements:

The project navigates the challenge of integrating a sensor system into biodegradable packaging, which is more expensive and mechanically sensitive than conventional plastics. By selecting PLA and designing a low-cost, low-power sensor with minimal footprint, the project demonstrates how conflicting requirements—affordability, functionality, and environmental compliance—can be reconciled.

P3 – Sensor Circuit Uncertainty:

The project explores a relaxation-oscillator temperature sensor as an alternative to conventional ring oscillators, voltage dividers, or other temperature-dependent circuits. By studying its temperature-frequency relationship under cold-storage conditions and calibrating against spoilage models, the design ensures reliable performance even in the presence of potential environmental variations such as minor humidity or oxygen exposure.

P4 – Food Safety and Biosafety Knowledge Gap:

While EE students typically lack formal training in food or biosafety, this project addresses these gaps by integrating food safety considerations into the sensor design and packaging. The mapping of temperature data to spoilage thresholds, adherence to safe storage bands for fruits and vegetables,

and isolation of electronics from direct food contact collectively ensure bio-compliance while educating students on cross-disciplinary problem-solving.

By fulfilling these objectives and systematically addressing the four CEPs, the project delivers a practical, scalable, and environmentally responsible smart packaging solution suitable for cold-storage applications in Bangladesh.

Literature Review and Material Selection

Literature Review

Smart food packaging has been the subject of extensive research as a means to address food safety and post-harvest losses by embedding sensor circuits capable of monitoring temperature, moisture, and gas levels. Studies have shown that such systems not only enhance shelf life but also improve traceability and sustainability when integrated with biodegradable materials. Research has particularly focused on the use of sensor circuits combined with modern simulation and layout tools such as SPICE to enable reliable performance analysis.

Previous studies on temperature monitoring have demonstrated that fluctuations accelerate microbial growth and spoilage, and therefore various sensor strategies have been proposed. Diffusion-based indicators, such as 3M Monitor Mark®, rely on the movement of colored substances to track cumulative heat exposure [1], while enzymatic indicators, such as VITSAB®, utilize lipid hydrolysis that causes measurable pH shifts [2]. More recent developments have explored IoT-enabled temperature sensors that transmit real-time data throughout the supply chain [3] and NFC-powered circuits that integrate wireless communication with temperature logging for cold chain monitoring [4]. In addition, studies have demonstrated that these temperature-sensitive circuits can be incorporated into biodegradable films such as chitosan and cellulose, combining functionality with environmental sustainability [5].

Research has also addressed the role of moisture in compromising food safety, as excess humidity accelerates mold and bacterial growth. Studies have investigated absorbent materials and electronic circuits for moisture detection, including systems like Thermoarite® that employ superabsorbent polymers to manage liquid exudate in poultry and fish packaging [6]. Capacitive and resistive circuits that measure dielectric or conductivity changes have been widely studied, with nanomaterial enhancements improving their sensitivity [7]. Additional work has proposed chitosan-based humidity indicators capable of colorimetric or resistance changes, particularly suited for dry food applications such as grains [8]. Integration with biodegradable substrates such as polylactic acid (PLA) has further been explored to support eco-friendly packaging suitable for low-income contexts [9].

Considerable attention has also been given to gas sensors as indicators of food spoilage. Studies have shown that oxygen sensors using photoluminescence quenching effectively track O₂ levels and help reduce oxidation in fruits and vegetables [10]. Other works have investigated reversible colorimetric indicators based on redox dyes such as methylene blue, capable of achieving low detection limits [11]. For carbon dioxide monitoring, sensors embedded in whey protein isolate films [12] and optical CO₂ sensors in bio-packaging [13] have been developed to provide visual spoilage alerts. Similarly, ammonia detection has been studied extensively, with TiO₂/MXene-based circuits offering high sensitivity [14] and curcumin-embedded polyethylene films providing visual feedback on NH₃ exposure [15]. Broader approaches using MOS sensors and electronic noses have demonstrated the detection of multiple volatile organic compounds, with machine learning applied for spoilage

prediction [16]. Additional studies have shown NFC-enabled paper-based circuits for ammonia detection in spinach packaging [17], biosensors utilizing biological elements for pathogen detection [18], and nanomaterials to improve selectivity. Such circuits have often been fabricated on cellulose or PLA substrates, aligning with sustainability goals [19], while low-cost printed electronics have been proposed to overcome cost and integration barriers in resource-constrained settings .

Overall, prior studies demonstrate that sensor circuits analyzed and optimized through SPICE simulations can be effectively integrated into biodegradable packaging materials to provide early and reliable indicators of food quality. These advancements underscore the potential of combining electronic sensing with sustainable packaging practices to enhance food preservation, safety, and environmental outcome.

Material Selection

In developing and integrating smart food packaging, packaging material selection is central in compromising among environmental sustainability, food safety, and sensor integration feasibility. For this project, Polylactic Acid (PLA) is selected as the biodegradable plastic substrate upon which the relaxation-oscillator-based temperature sensor would be integrated. PLA is a renewable resource-based thermoplastic sourced from corn starch or sugarcane, and it is completely biodegradable and compostable in industrial or domestic composting conditions. It is in compliance with environmental legislations, such as the Bangladesh Environmental Conservation Rules 2023(4), and aids in achieving the goal of reducing plastic waste in food packs.

PLA is thermally and mechanically stable enough to accommodate microelectronic devices such as the Schmitt trigger-based oscillator circuit without weakening the packaging structurally. Its non-toxicity and food-contact acceptability ratings ensure that the sensor is able to operate in close proximity to perishables such as fruits and vegetables without any possibility of contamination. Besides, PLA films are also compatible with lamination and thin-substrate embedding processing techniques, which enables efficient thermal coupling between the sensor and the packaged products and accurate temperature measurement.

Practically speaking, PLA is low-cost and easily available for deployment in low-income nations, enabling economies of scale for smart packaging solutions. Its printability, coating suitability, as well as for adhesives and protection, enables the addition of sensor electronics with a fully biodegradable package. PLA is hence a suitable solution to enable simultaneous solutions of food safety monitoring, green packaging, and compatibility with temperature-sensitive low-power electronics.

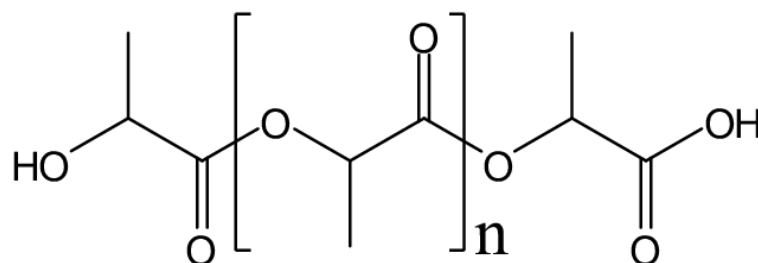
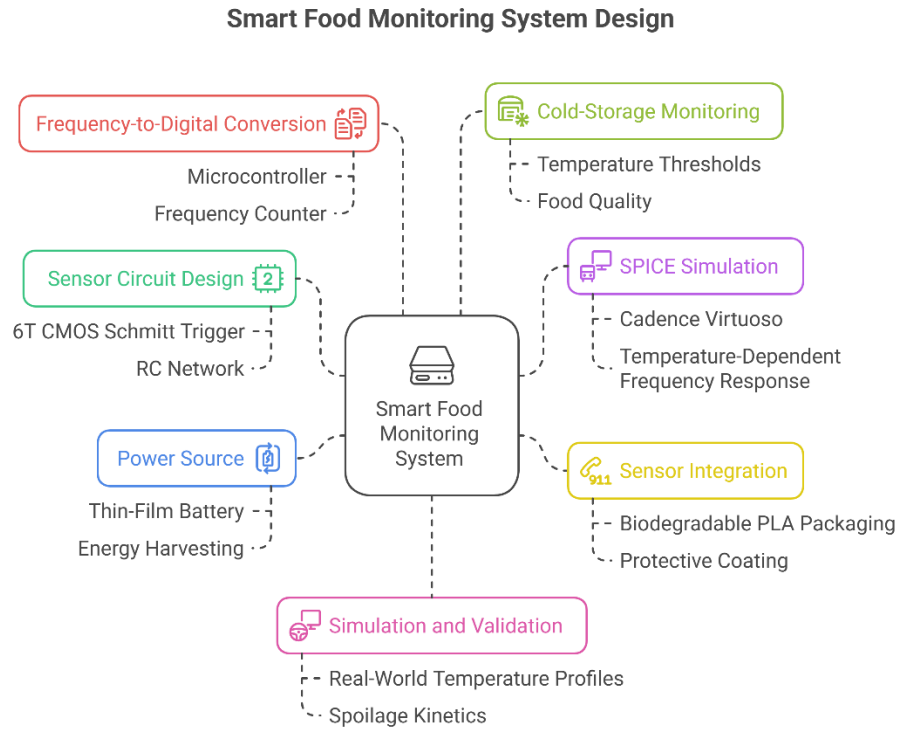


Fig: PLA Compound

Vision



Sensor Circuit Design & Tools Usage

A relaxation oscillator utilizing a 6-transistor (6T) CMOS non-inverting Schmitt trigger, coupled with an inverter, forms the core of a temperature-sensing system. This type of oscillator is often used in precision temperature measurement due to its simplicity and ease of integration. The frequency of oscillation generated by this configuration is sensitive to temperature variations, which can be exploited for temperature sensing applications. For the design and simulation of the necessary circuits, we have utilized **Cadence Virtuoso**, which allowed us to perform detailed **SPICE simulations** to verify the behavior and performance of the temperature-sensing system.

Basic Components and Circuit Configuration

Schmitt Trigger

Schmitt Trigger is a hysteretic comparator circuit, designed with an input that exhibits a **threshold hysteresis**. It has two distinct voltage thresholds: an upper threshold V_{UT} and a lower threshold V_{LT} . This hysteresis ensures that the output of the trigger does not change erratically due to noise or small fluctuations in the input signal. The **non-inverting Schmitt trigger** takes an input voltage and produces a clean digital output (either high or low) based on the state of the input voltage relative to the thresholds V_{UT} and V_{LT} .

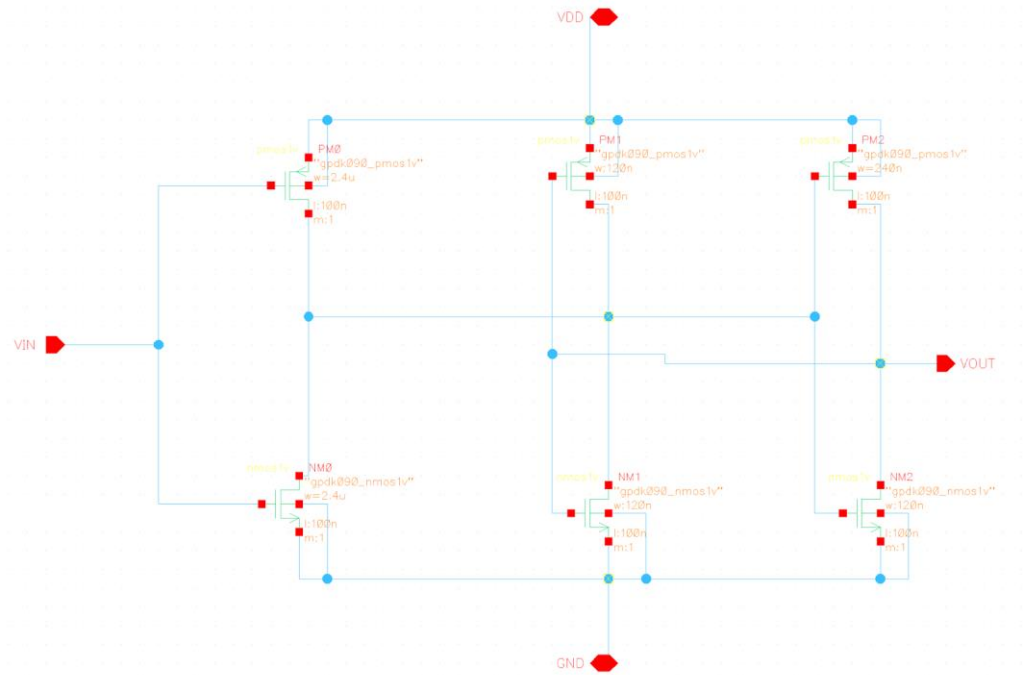


Fig: Schmitt Trigger Circuit.

Inverter

An **inverter** is connected to the output of the Schmitt trigger to create an **inverting Schmitt trigger configuration**. This ensures that the output alternates between two voltage levels, typically V_{OH} (high voltage) and V_{OL} (low voltage).

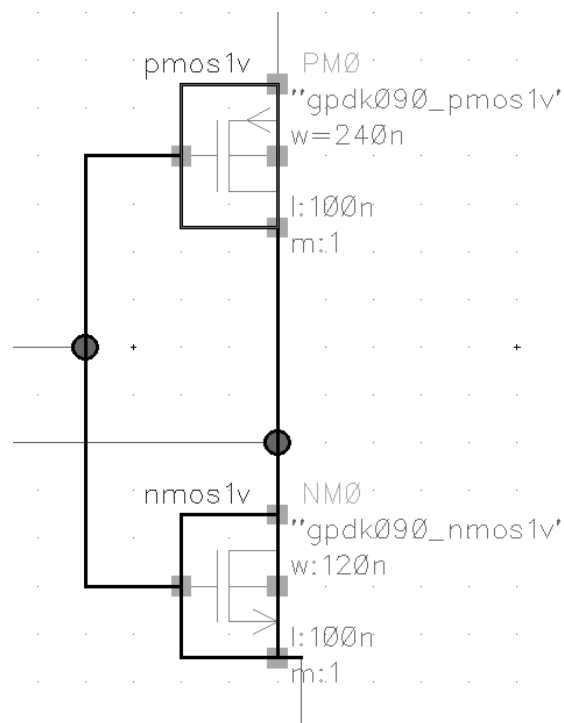


Fig: Inverter Circuit

RC Network

A **passive RC network** consisting of a resistor R_R and an effective capacitance C_{eff} is connected between the output of the Schmitt trigger and its input. This RC network is responsible for generating the oscillations.

The capacitor C_{eff} charges and discharges through the resistor R_r , and the voltage across the capacitor determines the state of the Schmitt trigger's input, causing it to flip between high and low states, generating a square wave output.

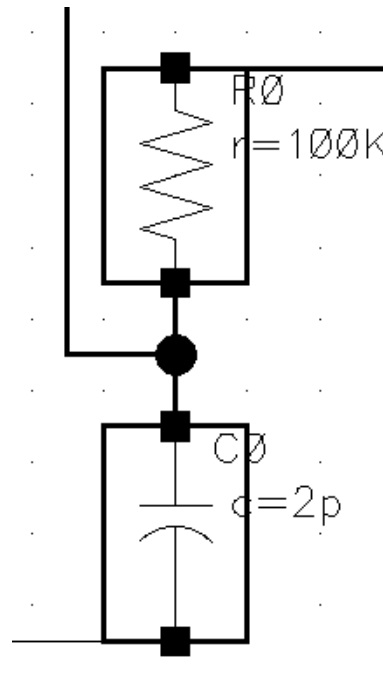


Fig: RC Network Circuit

Charge-Discharge Cycle

The core operation of the oscillator relies on the periodic charging and discharging of the capacitor C_{eff} through the resistor R . The voltage across the capacitor changes in a characteristic way, which governs the timing of the oscillations:

When the output is high (V_{OH}): The capacitor C_{eff} begins to charge towards the high voltage level V_{OH} . The rate of charging depends on the RC time constant $\tau = R C_{eff}$. The voltage at the capacitor's input increases exponentially.

$$\tau = R.C_{eff}$$

Threshold Crossing (V_{UT}): Once the voltage across the capacitor exceeds the upper threshold V_{UT} , the Schmitt trigger switches its output from high to low.

When the output is low (V_{OL}): The capacitor now discharges towards V_{OL} . The discharge follows an exponential decay governed by the RC time constant.

Threshold Crossing (V_{LT}): When the capacitor voltage drops below the lower threshold V_{LT} , the Schmitt trigger switches back to the high state.

Charging time:

$$t_{\text{charge}} = RC_{\text{eff}} \ln \left(\frac{V_{OH} - V_{LT}}{V_{OH} - V_{UT}} \right)$$

Discharging time:

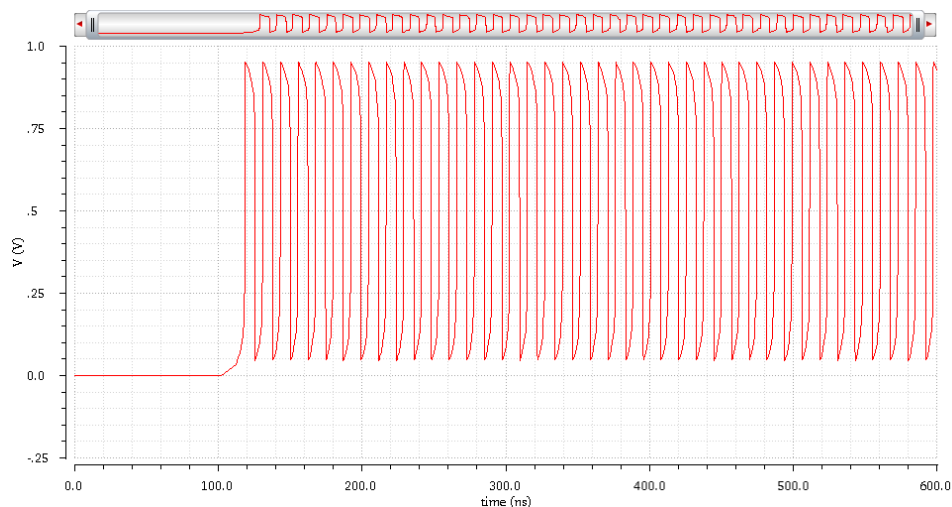
$$t_{\text{discharge}} = RC_{\text{eff}} \ln \left(\frac{V_{UT} - V_{OL}}{V_{LT} - V_{OL}} \right)$$

Total period and frequency:

$$T = t_{\text{charge}} + t_{\text{discharge}}, \quad f = \frac{1}{T}$$

This charging-discharging cycle repeats, producing a square wave output signal with a frequency determined by the charging and discharging time of the capacitor.

Transient Analysis plot for a temperature:



Temperature Sensor Implementation

In the temperature-sensing implementation, the relaxation oscillator acts as a transducer, where temperature affects the electrical quantities that influence the oscillation period, such as $R(T)$, $C_{\text{eff}}(T)$, the Schmitt trigger thresholds $V_{UT}(T)/V_{LT}(T)$, and the output levels $V_{OH}(T)/V_{OL}(T)$. This relationship results in a temperature-dependent frequency, denoted as $f(T)$

The frequency can be approximated by the relation:

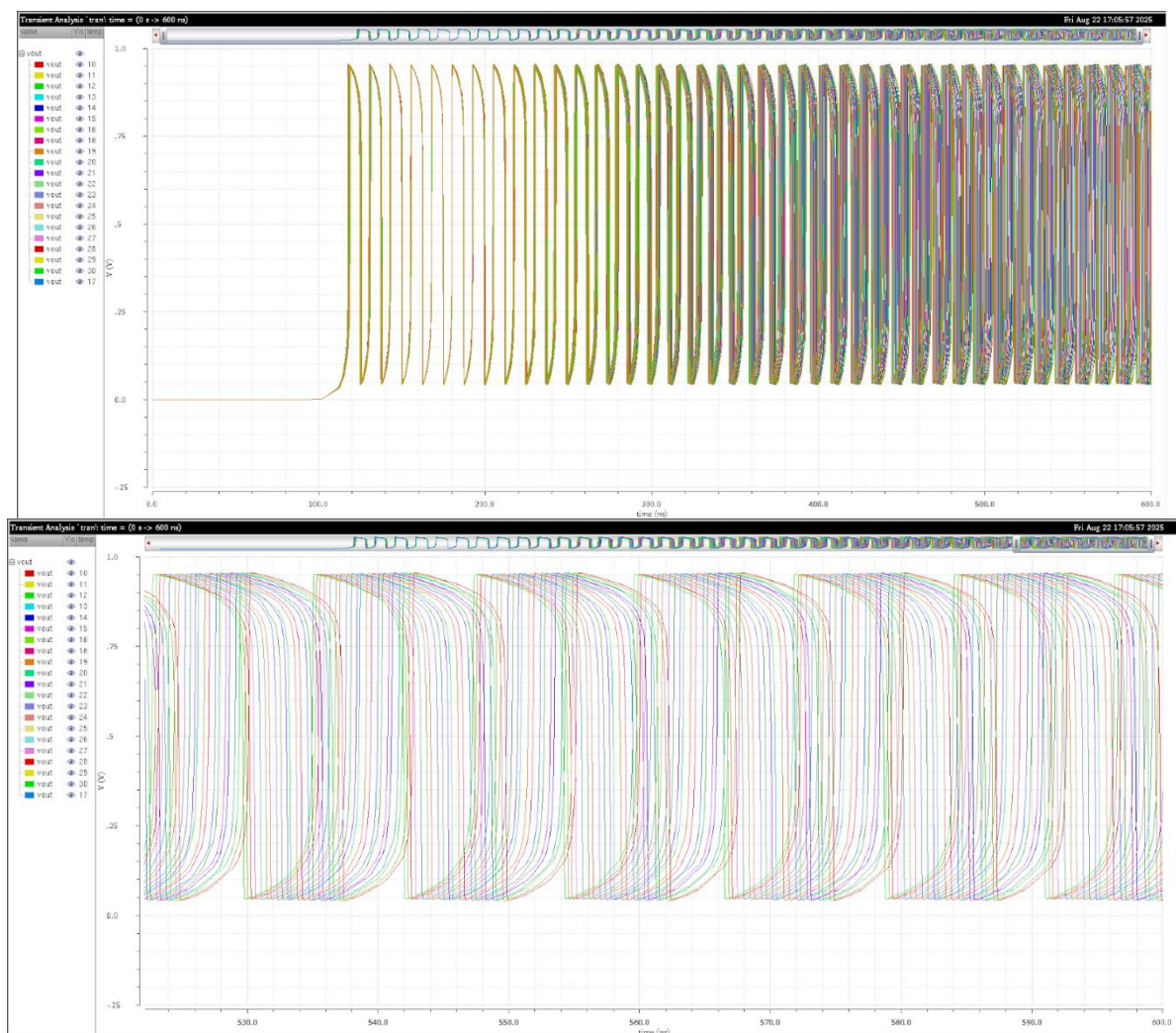
$$f(T) \approx \frac{K(T)}{R(T)C_{\text{eff}}(T)}$$

Where $K(T)$ represents a threshold/output logarithmic factor. Linearizing the equation for small temperature changes gives the following normalized sensitivity equation:

$$\frac{1}{f} \frac{df}{dT} = \frac{1}{K} \frac{dK}{dT} - \frac{1}{R} \frac{dR}{dT} - \frac{1}{C_{\text{eff}}} \frac{dC_{\text{eff}}}{dT}$$

This shows how the temperature variations in the system influence the frequency. If the measured frequency shows a monotonic, approximately **linear** increase with temperature, it suggests that the combined effects of temperature-dependent changes in the Schmitt thresholds and output levels $K(T)$ dominate over the temperature dependencies of the RC product $R(T)C_{\text{eff}}(T)$.

Output Curve from SPICE simulation using Cadence:

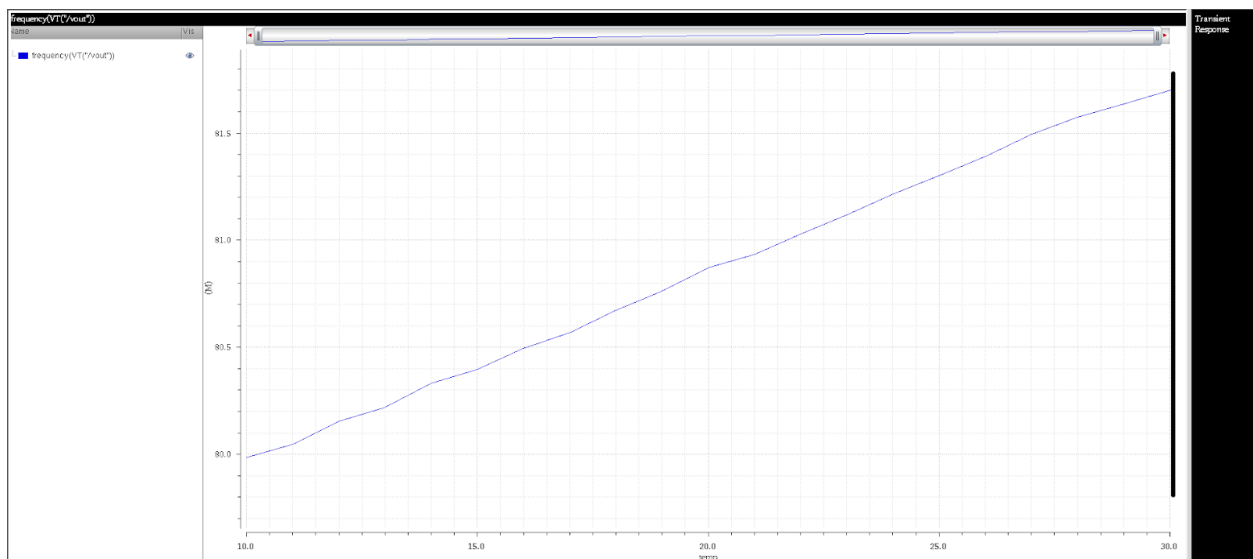


Linearization and Sensitivity

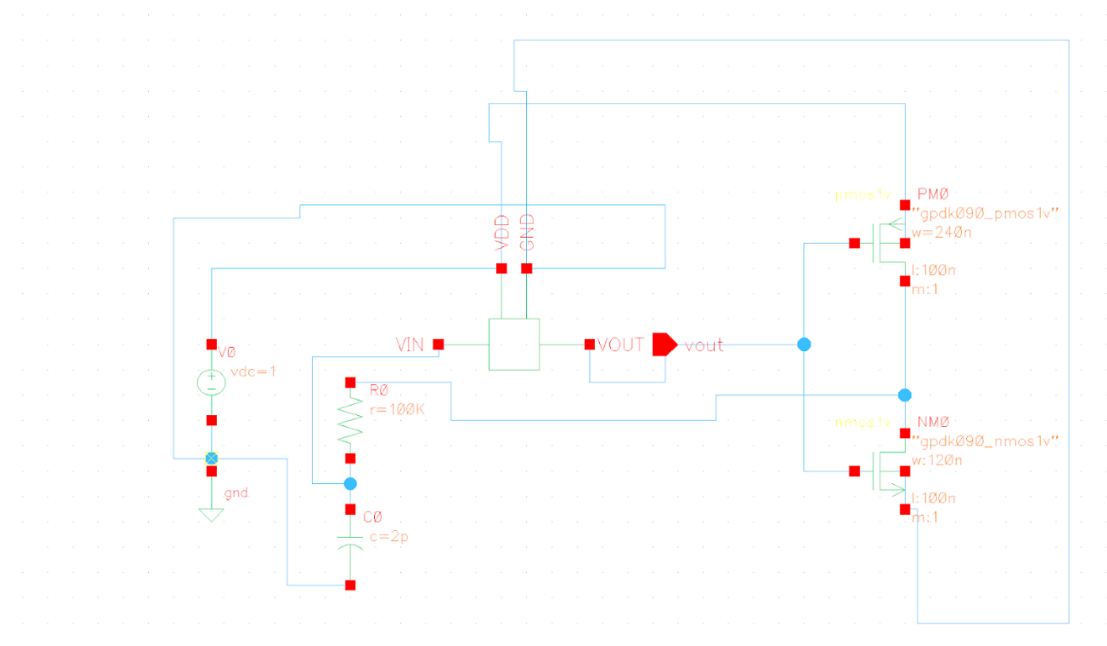
For small temperature variations, the frequency-temperature relationship can be approximated as nearly linear. This can be useful in temperature sensing applications where a **linear output** is desirable.

A **linear increase** in frequency with temperature indicates that the combination of threshold voltage and output level changes has a stronger effect than the temperature dependence of the RC product $R(T) \cdot C_{eff}(T)$.

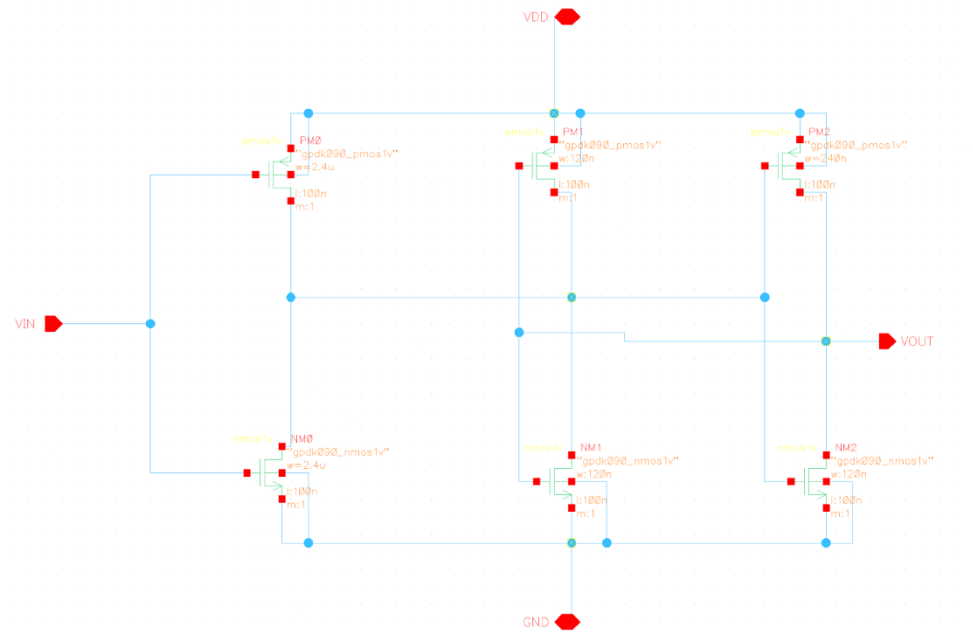
Simulation output (Temperature vs Frequency):



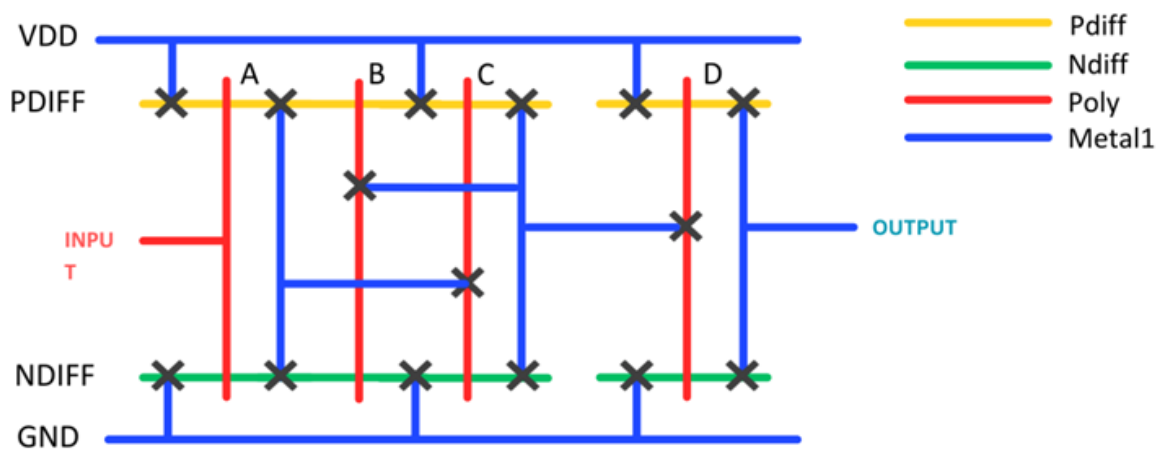
Complete Circuit Diagram:



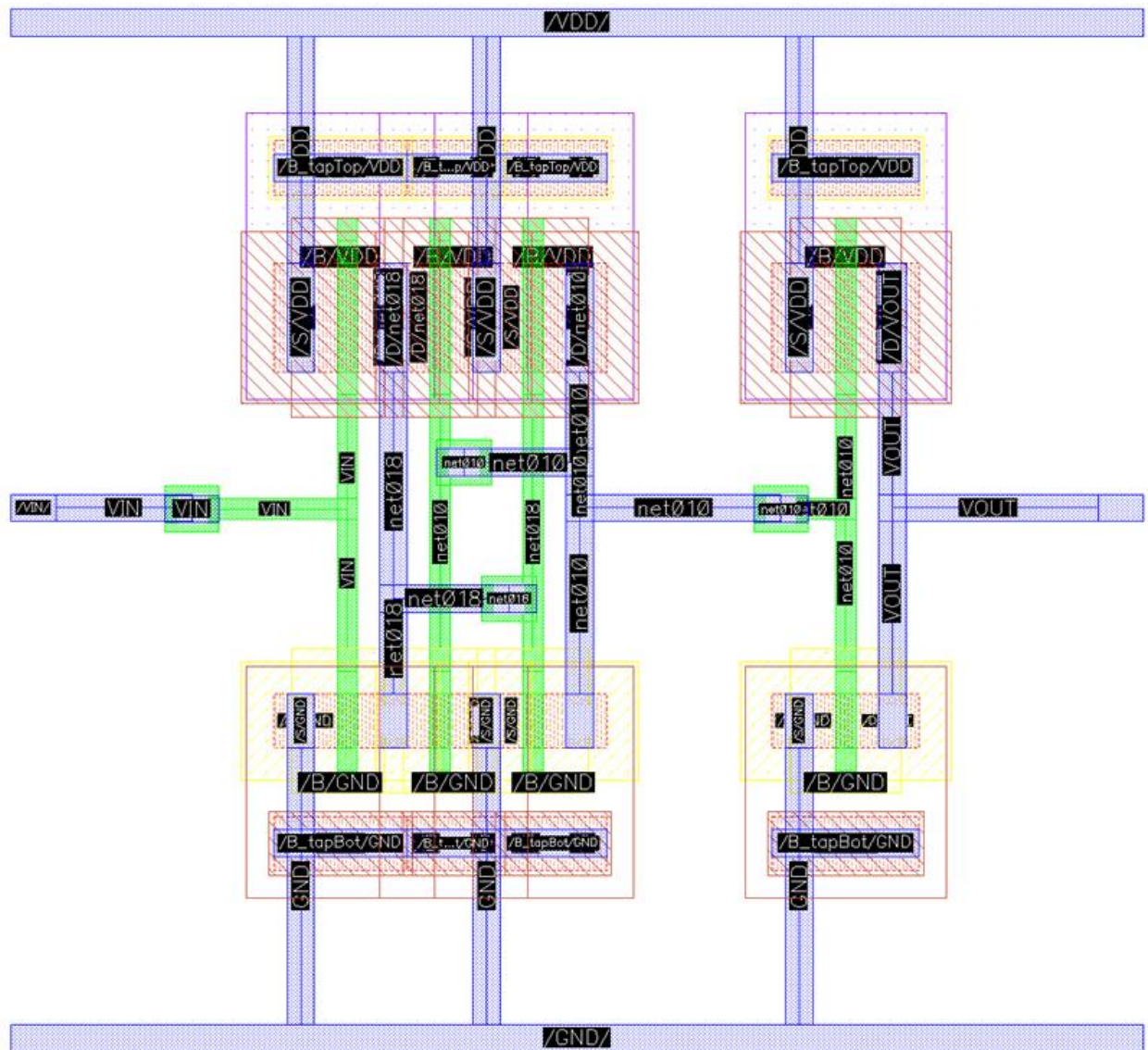
Schmitt Trigger Circuit:



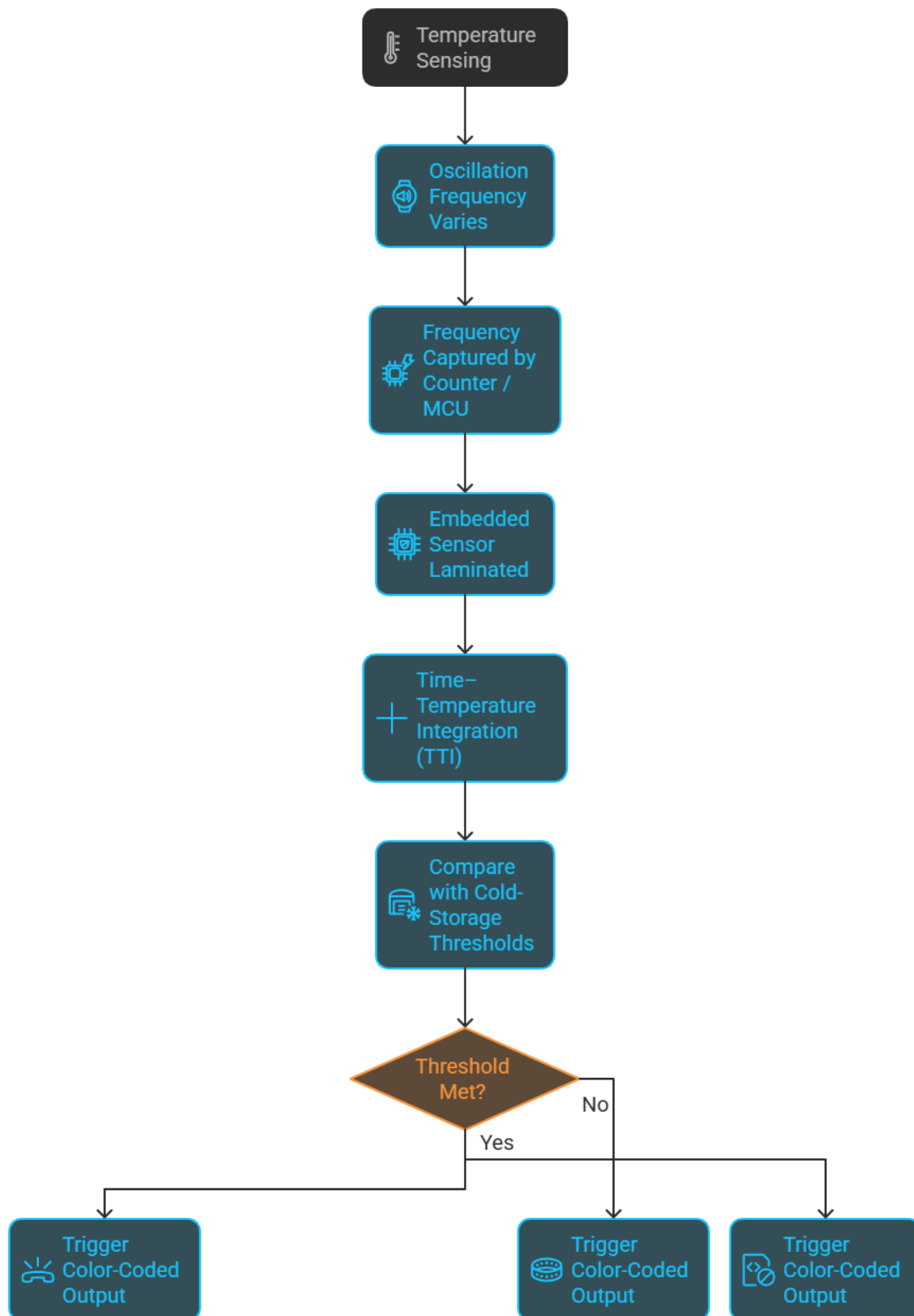
Stick Diagram:



Layout:



Summary Workflow:



Integration with Packaging

Sensor Embedding and Functional Design

The temperature-sensing component developed is to be embedded in PLA (polylactic acid) packaging to make the package active in contributing towards cold-chain preservation. The oscillator circuitry (a 6-transistor CMOS non-inverting Schmitt trigger cascaded with an inverter) is fabricated on a thin, flexible substrate and laminated on the inside surface of the PLA pouch or tray wall to expose it to package headspace/ambient temperature to be detected while physically isolated from direct contact with food by a thin food-grade protective coating. A thermally conducting adhesive layer is used to promote thermal coupling and prevent lag response without compromising the package barrier. The oscillator temperature-varying frequency is linked to a miniature frequency-to-digital stage (an ultra-low-power counter or small MCU), which computes a time–temperature integral (TTI) from repeated frequency readings and employs a calibrated lookup to produce a freshness index.

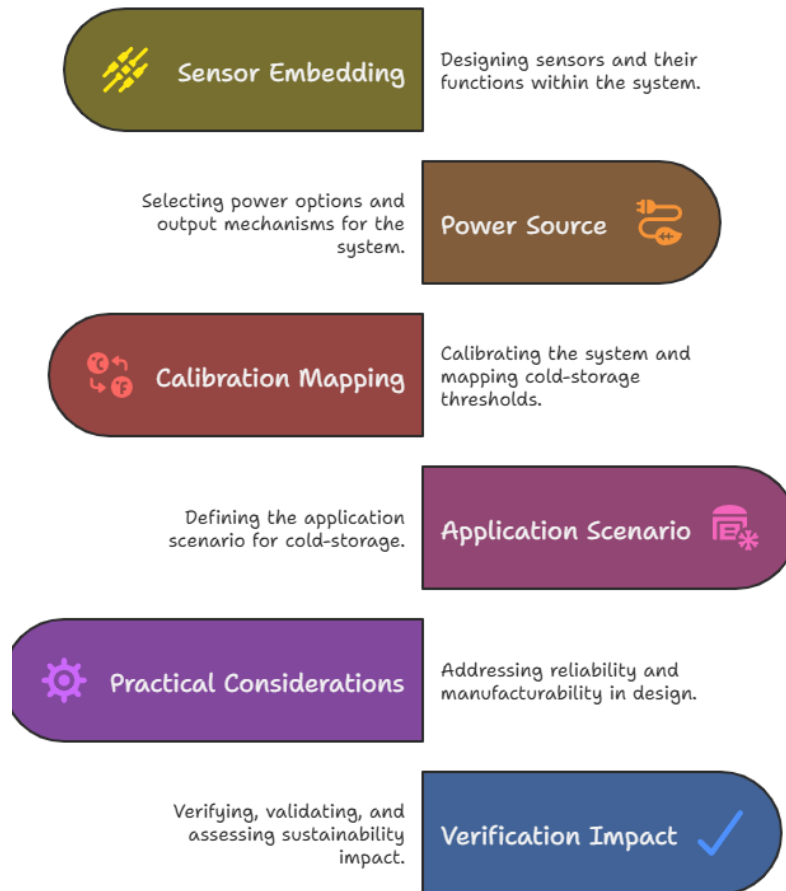
Power Source Options and Output Mechanism

Two power practice solutions are proposed for meeting different deployment needs. A small printed coin cell or thin-film battery of corresponding size to expected shelf life (weeks) may be used for disposable retail packaging; the electronics are duty-cycled (sleep in between short sampling bursts) in an effort to keep average current levels within microampere ranges. For reusable or logistics-stage tags, passive or semi-passive energy scavenging (NFC/RF harvesting during checkpoint readings, or miniaturized solar/thermoelectric harvesters to accumulate energy during the day) is suggested to preserve battery life and extend useful lifetime. Hardware-only threshold comparators can be used to trigger a tri-color LED or a thermochromic printed marker for purely self-contained visual display, and an MCU option is more flexible in accumulation of exposure minutes and commodity-dependent calibration.

Threshold mapping and calibration with cold storage

Calibration is done by modeling the oscillator frequency as a function of temperature in a controlled chamber over the range of cold storage. The frequency–temperature mapping derived from this is combined with spoilage kinetics (e.g., Q10 or Arrhenius relationships and commodity shelf-life data) to lead to a simple TTI policy that is implemented on the device. For mixed fresh fruits and vegetables under standard cold-chain conditions, working thresholds are set to represent cold-storage safety: Safe corresponds to continuous exposure between 0–4 °C; Warning is initiated when cumulative exposure between >4–8 °C exceeds roughly 2 hours within any 24-hour block of time; Unsafe is signaled if exposure exceeds >8 °C for more than 1 hour or if any transient surge above 10 °C is detected. These bands are adjustable by SKU (e.g., reduced target ranges for leafy greens and increased tolerances for fruit tolerant to chill) through adjustment of the stored look-up table in the converter/MCU.

INTEGRATION WITH PACKAGING FLOWCHART



Cold-Storage Application Scenario

A typical cold-storage scenario illustrates the basis of this approach: pallets being delivered to a cold room in a distribution center at $\sim 2^{\circ}\text{C}$ are monitored in real time; door-openings or handling over short time-intervals cause temporary increases in temperature that the TTI logic cumulates within the warning band, so that only packages accumulating large quantities of higher-temperature time are marked. If a refrigeration or transport failure results in a several-hour increase to $8\text{--}9^{\circ}\text{C}$, off-spec packages are labeled as unsafe on delivery and might be quarantined or queued for immediate sale. This time-based, frequency-sensitive detection thereby also supports operational decisions such as FEFO (first-expired, first-out) treatment and both reduces health risk and undesirable waste.

Practical concerns with Reliability and Manufacturability

Certain engineering details must be overcome for successful operation. Condensation and moisture must be prevented by conformal overcoats and edge sealing to avoid long-term drift of the RC network; thermal lag is avoided by the sensor proximity to the inner wall and the use of thin adhesives; sampling intervals and duty cycles are calibrated for battery life without sacrificing early detection (e.g., minute-

level sampling with deep sleep between samples). Manufacturability is retained through packaging of the sensor electronics in label-type insert that can be fitted inline at filling, with upstream packaging processes remaining unchanged and allowing for removal of sensors before composting as needed. End-of-life processing is managed such that the PLA package remains compostable while removable electronic parts are recovered in line with local recycling channels.

Verification, Validation, and Sustainability Impact

They are executed using chamber step/soak tests, pallet tests with comparative data loggers, and shelf-life correlation experiments (microbial analyses, sensory evaluations) to ascertain that the TTI thresholds reflect meaningful quality losses. Through these provisions, the suggested relaxation-oscillator sensor integrated in PLA packaging provides an affordable, energy-saving, and convenient means of ongoing cold-storage monitoring that directly contributes to rot prevention in fruits and vegetables while remaining compatible with environmental sustainability objectives.



Fig: Product Prototype

Decision-Making Algorithm/ Framework

This systematic decision-making approach seeks to balance food safety, human health, and sustainability in food packaging in Bangladesh in line with the Bangladesh Environmental Conservation Rules 2023. The combination of cutting-edge sensor technology, biodegradable packaging materials, and a holistic approach to sustainability supports both consumer safety and the ultimate aims of minimizing food waste and plastic pollution.

1. Food Safety

Ensure the food being supplied to consumers is safe and free from contaminants due to spoilage, which in most instances is caused by improper storage and inadequate temperature control in the process of distribution.

Real-Time Temperature Monitoring

mount smart packages with in-built temperature-sensitive sensors like the relaxation oscillator-based sensor in the project. The sensors will monitor the food storage conditions in real-time, ensuring the food remains safe within the acceptable temperature range to prevent microbial growth and spoilage. The sensor senses frequency change with temperature, providing a real-time freshness index (Green: Safe, Yellow: Warning, Red: Unsafe).

Tracking food conditions (e.g., time and temperature exposure) will allow consumers and retailers to quickly determine if the food was stored under unsafe conditions. Such an upfront monitoring system significantly reduces the likelihood of eating spoiled or unsafe food.

Integration with Cold Chain Management:

Cold chain is necessary for the storage of perishable food. This smart packing solution integrates into the cold chain network such that temperature exposure monitoring is easy and such food stuffs as meat, milk, and vegetables are stored and transported in optimum condition. The Bangladesh Food Safety Standards (storage conditions and temperatures) must be adhered to using this technology.

Regulatory Compliance: The sensor system will be in compliance with Bangladesh's food safety regulations, allowing for safe procedures to be followed by all the stakeholders, from producers to retailers. Additionally, the speed of the sensor can be calibrated to food spoilage models (for instance, the Q10 or Arrhenius models) to provide accurate judgments of food safety based on exposure to temperature.

2. Human Well-being

Preserve the consumer's health by reducing food intake that is spoiled or unsafe, and simultaneously inform the public of food quality and safety through intelligent package systems.

Prevention of Foodborne Illness

Intelligent packaging informs consumers of inappropriate food storage conditions. By constant monitoring of temperature and the provision of a direct warning in color-coded form (Green: Safe, Yellow: Warning, Red: Unsafe), the system reduces the risk of foodborne illness from inappropriately stored or expired products.

Waste Reduction

When food is stored at temperatures above safe temperatures for prolonged periods, the sensor system provides alerts that trigger the consumption or destruction of food in advance before the food gets dangerous, hence minimizing food waste. This minimization of food wastage not only benefits

consumers but also increases food security by way of delivering food to the market while still fresh and consumable.

Increased Consumer Confidence:

With the smart, sensor-embedded packaging, consumers are made aware of the state of their food, giving them confidence in the safety and freshness of what they purchase. Such openness can alter the way consumers act and drive consumers toward healthier food, as individuals become more aware of the goodness of storing food properly.

Regulatory Compliance for Consumer Health

Compliance with the food safety standards of the local authority is the top priority in maintaining human wellbeing. The packaging solution meets Bangladesh's food safety and sanitation standards that preclude foodborne illness from spoilage or contamination.

3. Sustainability

Promote sustainability of the environment by reducing plastic use and energy usage through low-power sensor technology and biodegradable materials.

Biodegradable Packaging (PLA):

One of the major ways in which this framework is directed towards sustainability is in the use of Polylactic Acid (PLA) in packaging. PLA is manufactured from renewable resources such as corn starch or sugarcane, and it is biodegradable, thereby reducing plastic litter. This aligns with the Bangladesh Environmental Conservation Rules 2023, which mandate the reduction of non-biodegradable waste. The packaging is not harmful, thereby being safe for use by consumers, yet still being eco-friendly.

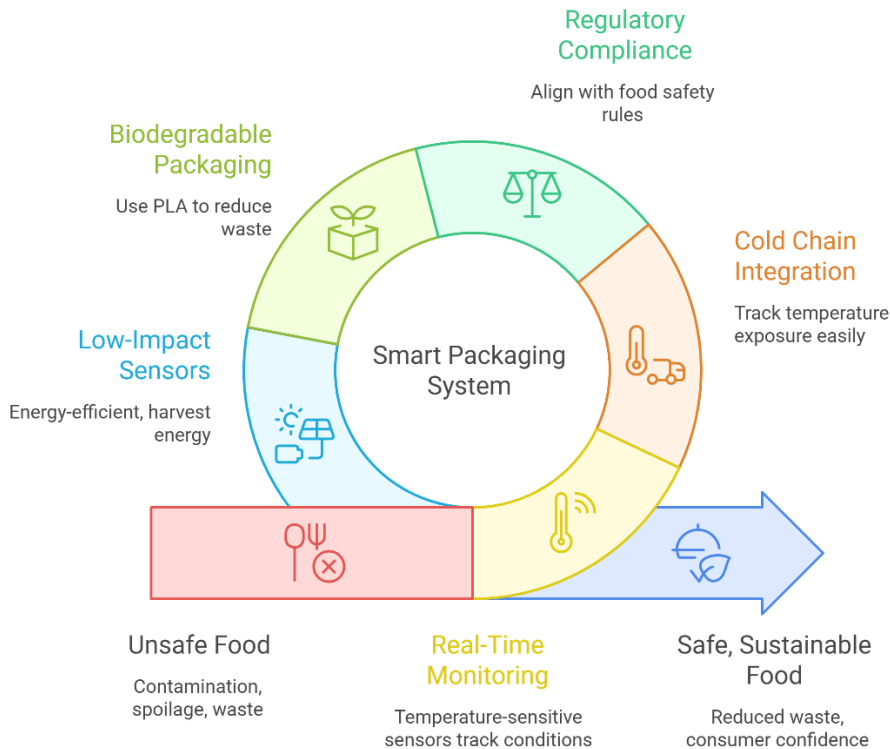
PLA packaging is easier to degrade than the conventional plastics, which minimizes landfill waste and assists in realizing Bangladesh's Zero Waste Movement goals.

Low-Impact Sensors:

Use of a low-power relaxation oscillator sensor, with less energy consumption, will render the system energy efficient. Additionally, the energy source for the sensor can be transferred via energy-harvesting technologies (e.g., RF or NFC power), reducing dependence on batteries and minimizing environmental footprint. The packaging system, through embedded energy-harvesting technology, not only becomes environmentally friendly but also complies with Bangladesh's growing enthusiasm for green energy technology. This reduces the carbon footprint of each package that has sensors, supporting the climate change mitigation efforts of Bangladesh. Energy-Efficient Manufacturing

The proposed system utilizes locally accessible components that are inexpensive and widely available, making the solution scalable and feasible for large-scale manufacture. Moreover, the PLA packaging material is widely utilized in Bangladesh's food packaging industry, which facilitates the existing infrastructure for sustainable food packaging solutions.

Smart Packaging for Food Safety



Alignment with Bangladesh Environmental Conservation Rules 2023

The proposed model of biodegradable smart food packaging is in direct compliance with Bangladesh's Environmental Conservation Rules 2023, which emphasizes sustainability, waste minimization, and compliance with national food safety standards. The use of Polylactic Acid (PLA) as the primary packaging material, coupled with low-power temperature-sensing circuits, is in compliance with several national goals of minimizing plastic waste, promoting the use of sustainable materials, and ensuring food safety. The technical modalities of how this solution meets these regulatory conditions are explained in the following sections.

1. Reduction of Plastic Waste

PLA as a Biodegradable Packaging Material:

PLA is a biodegradable thermoplastic polymer made from renewable resources like corn starch or sugarcane. PLA, in contrast to traditional petroleum-based plastics, breaks down under composting conditions, and thereby it reduces the environmental pollution from plastic waste to a great extent. This aligns with Bangladesh's policy of reducing the use of non-biodegradable plastics.

Technical Integration: PLA substrate is thermally stable and mechanically strong to hold the sensor circuit (relaxation oscillator) safely without compromising food safety and integrity. Compatibility of PLA with printed and embedded electronics enables sensors to be embedded directly into the packaging material. Biodegradability of PLA ensures that after use, the packaging will naturally break down without contributing to long-term landfill expansion.

2. Environmentally Friendly Materials Promotion

Biodegradable Sensors Integrated into Packaging:

The sensor design takes advantage of low-power electronics to ensure low power consumption and enables environmentally friendly operation. The temperature sensor fabricated based on a 6T CMOS Schmitt trigger oscillator works efficiently within the low-power constraint of the biodegradable PLA substrate to reduce the overall environmental footprint.

Technical Integration: The sensor employs low current consumption (in microampere range) through duty-cycling the system (sleeping between active sensing intervals). This minimizes the power consumption, which can be powered by energy-harvesting solutions (for example, NFC or RF harvesting). This minimizes the utilization of batteries, again being consistent with the environmental goals of decreasing the number of disposable power supplies used in food packages.

Moreover, integrating the sensors in the packaging itself eliminates the need for external monitoring devices, thus reducing packaging material waste in the form of additional sensors and labels traditionally used in food packaging.

3. Sustainable Manufacturing

Local Sourcing and Low-Cost Materials:

The use of locally sourced PLA and low-power, low-cost components enables the smart packaging system to be low-cost and scalable for use in Bangladesh. It helps to support the local industry as well as reduces the carbon footprint associated with shipping high-value non-renewable materials.

Technical Manufacturing Consideration: The flexible PCB (Printed Circuit Board) used for sensor integration can be manufactured locally using low-cost equipment and materials, reducing both the carbon footprint as well as the cost of manufacturing. Using low-cost sensors and low-power consumption components (e.g., thin-film batteries or RF energy harvesting), this solution can be offered to small-scale producers and local food producers at low cost, enabling sustainable practices for local industries.

The simplicity of the relaxation oscillator circuit design (with easily available 6-transistor CMOS technology) minimizes the requirement of advanced components, thus supporting low-cost, large-scale manufacturability within national sustainability plans.

4. Compliance with Food Safety Standards

Real-time Food Safety Monitoring:

The integration of the temperature sensor into biodegradable PLA packaging offers continuous, real-

time food storage condition monitoring. This allows stakeholders (producers, distributors, and retailers) to track temperature excursion that may compromise food safety, ensuring compliance with Bangladesh Food Safety Standards.

Technical Functionality: The relaxation oscillator-based temperature sensor tracks changes in ambient temperature inside the packaging and gives an output frequency in direct proportion to the ambient temperature. This electronic output from frequency is mapped to food spoilage models (e.g., Q10 or Arrhenius models) to facilitate precision food quality predictions from time-temperature exposure. The sensor-based packaging system facilitates traceability by recording exposure to sub-optimal conditions, such as temperature excursions beyond safe levels for perishable products like meat, dairy, and fruits. This provides an accountable and transparent cold chain that is needed for both global regulatory compliance and national food safety regulations.

Regulatory Compliance: The smart food packaging system complies with the food safety laws of Bangladesh because it maintains food within safe temperature ranges and also warns if there is any deviation. The color-coded freshness indicators (Green, Yellow, Red) provide simple-to-understand information to food handlers and consumers so that food safety decisions are made beforehand, based on real-time data.

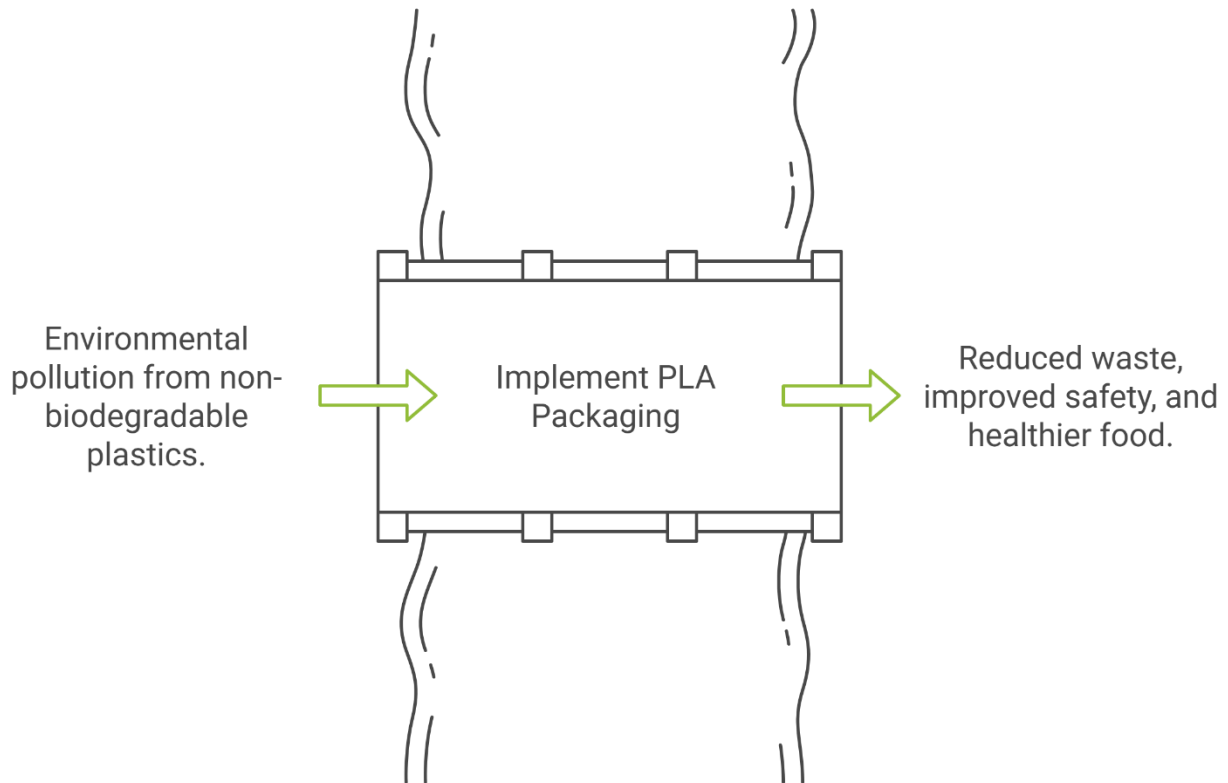
5. Contribution to a Healthier Food System

Improved Food Safety: By continuously tracking the storage conditions, the system ensures that temperature-sensitive foods are kept in safe temperatures. This reduces the chances of foodborne illness caused by improper storage or spoilage.

Technical Monitoring: The sensor system provides a freshness index that can be used to identify unsafe food products before they reach the consumers. The ability of the system to track the exposure of foods to high temperatures (e.g., above 4°C) for extended periods, and label foods as unsafe when required, ensures public health safety and discourages the consumption of tainted food.

Food Waste Reduction: The ability to detect potential spoilage ahead of time helps stores and consumers make timely food use decisions, thus reducing wastage and guaranteeing food safety.

Biodegradable smart food packaging ensures food safety and sustainability in Bangladesh.



Proposal Quality and Originality

Novelty and Unique Features

The novelty in this proposal lies in the merging of a low-cost oscillator circuit with VLSI and biodegradable PLA packaging to create an intelligent food safety indicator. In contrast to the conventional packaging with barcodes, diffusion-based indicators, or big IoT sensors, the approach in question integrates a 6-transistor CMOS Schmitt trigger–inverter relaxation oscillator with packaging films in a direct manner. The circuit offers an oscillation frequency based on temperature, which is correlated against microorganism model spoilages such as Q10 and Arrhenius kinetics for real-time freshness detection. This is technologically novel as it translates an analog electrical parameter (oscillator frequency) to a digital freshness score, bridging circuit-level design to food preservation. Furthermore, PLA-based biodegradable substrate usage complies with Bangladesh Environmental Conservation Rules 2023, with maximum environmental sustainability ensured with consumer safety.

Technical Depth and Innovation at the Circuit Level

The proposal exemplifies robust technical depth through the grounding of design in system integration and device physics. The principle of working of the oscillator is mathematically represented using RC charging–discharging dynamics and hysteresis thresholds (V_{UT} and V_{LT}). The charging/discharging time constants are represented as logarithmic functions of voltage levels such that the oscillation frequency $f=1/T$ can be derived in closed form. For enhanced precision, frequency-temperature relationship is measured by the sensitivity model linearized $(\Delta f/f_0)(1/\Delta T)$. Such level of circuit analysis ensures that performance becomes predictable, simulatable, and verifiable. Furthermore, the use of Cadence SPICE simulation, stick diagrams, and layout design helps to highlight the adoption of modern EDA (Electronic Design Automation) tools in meeting PO5 requirements for engineering practice.

Feasibility and Implementation Quality

The proposed relaxation-oscillator sensor in PLA packaging is viable, cost-effective, and environmentally friendly to Bangladesh. The structure is technologically uncomplicated, rugged, and energy-conserving with minimal components and accurate monitoring of fruits and vegetable cold-storage conditions. In terms of operation, the color-coded freshness indicator (Green/Yellow/Red) provides intelligent guidance without needing professional interpretation.

It is affordable in the Bangladeshi environment, being highly cost-efficient with locally available or low-cost imported materials. Mass production and low-cost PLA packaging also render it extremely possible for mass-scale deployment.

Component	Quantity	Unit Cost (BDT)	Total Cost (BDT)
6T CMOS IC / Transistors	1	50	50
Resistors & Capacitors	3	2	6
Flexible PCB Substrate	1	100	100
Microcontroller	1	50	50
Battery / Thin-film power source	1	50	50

PLA Packaging	1	50	50
Protective Coating / Adhesive	1	20	20
Tri-color LED / Thermochromic Label	1	10	10
Total	-	-	336

Table: Financial Breakdown.

Environmental-wise, PLA encourages biodegradability and compliance with Bangladesh Environmental Conservation Rules 2023(4). Low-power sensor encourages potential energy-harvesting methods, reducing environmental impact even further. Estimated at 336 BDT per packaged sensor, this is within affordable limits for cold-storage application in Bangladesh with assured effective monitoring of food quality, reducing post-harvest losses, and encouraging sustainable practices.

Limitations and Acknowledgement of Challenges

The proposal is self-critical, and this makes it stronger. It acknowledges that spoilage detection is indirect from time–temperature history rather than direct microbial measurement. Relying on calibration for different food classes (meat, dairy, fruit) introduces variability, and oscillator drift caused by humidity, mechanical stress, or material aging could affect accuracy. PLA's short shelf-life and the challenge of having a stable power source are other limitations. At the engineering level, it is difficult to integrate thin-film electronics into biodegradable substrates in terms of integration and miniaturization. Furthermore, environmental instability (humidity, uneven cooling) and local regulation (FDA, EFSA) also make it difficult to scale this system globally.

Proposal Quality and Logical Flow

The proposal has clearly defined and logical order, beginning with the background of food safety and sustainability concerns, going through literature review, theoretical modeling, circuit design and tool use, and concluding with integration into PLA packaging. It is trailed by characteristic features, feasibility, constraints, and challenges, making it complete and consistent. This trajectory highlights the engineering creativity (oscillator design, calibration, sensor-packaging integration) and the social relevance (sustainability, health, waste reduction). The balance between novelty, technicality, feasibility, and limitations validates the general quality of the proposal and renders it an academically innovative and practicable solution.

Conclusion

The proposed biodegradable smart packaging system for food marks a significant innovation in the monitoring of food quality across the supply chain, particularly in Bangladesh. With the integration of a temperature-sensitive relaxation oscillator within PLA (polylactic acid) packaging, real-time monitoring of temperature fluctuations is made possible, which are critical in maintaining the freshness and safety of perishable food items like fruits and vegetables. The oscillator, which is an oscillator based on a 6-transistor CMOS Schmitt trigger circuit, alters its oscillation frequency with temperature variation, and this is translated into a digital freshness index. This allows stakeholders to track temperature excursions that would impact food quality, generating pre-emptive warnings that can induce timely action to prevent spoilage.

One of the major advantages of this approach is its compatibility with sustainability goals. PLA, the packaging material, is biodegradable and compostable, addressing the environmental issue posed by conventional plastic packaging directly. The use of PLA is consistent with Bangladesh's Environmental Conservation Rules 2023, which aims to reduce plastic waste and encourage the use of eco-friendly material. By combining an energy-saving sensor with a biodegradable matrix, the solution not only enhances food safety but also contributes to the global struggle against plastic pollution.

In addition to the primary goal of food safety, the proposed system enhances traceability and transparency within the food supply chain. By embedding the sensor in the packaging, the system ensures that the monitoring process is ongoing and intrinsic to the packaging, reducing the need for additional sensors or labels, ultimately saving waste of material. The freshness index, represented by a straightforward color-coded scheme (Green Safe, Yellow Warning, and Red Unsafe), provides a simple-to-interpret system in which consumers and retailers can easily make well-informed decisions about the food they are handling and so promote healthier food choices and reduced foodborne disease.

Although there is high potential for food safety, sustainability, and operational effectiveness in the system, several barriers exist. Temperature sensor calibration versus different food types and potential environmental sensor drift caused by humidity or mechanical stress needs to be addressed to ensure long-term reliability and accuracy. Additionally, the low shelf life of PLA, as well as the difficulties of maintaining a steady power source for the sensor, may be challenges to broad-scale deployment. But these barriers can be addressed by additional design development, e.g., by integrating energy-harvesting technology (e.g., NFC or RF harvesting) that reduces battery reliance.

Despite these barriers, the system here is a practical, scalable, and cost-effective solution to Bangladesh's environmental and food safety issues. Its low price, low energy needs, and minimalist design guarantee its robust compatibility with the local environment, where there is a need for an affordable food packaging system. Furthermore, the system's compliance with Bangladesh food safety standards guarantees that it satisfies national requirements, and both provides a consistent method of food safety monitoring and a quantifiable contribution to environmental preservation.

In short, this smart food packaging system is an efficient solution to augmenting food storage and distribution processes in Bangladesh, particularly where post-harvest loss and food-borne illness are prevalent. Addressing the dual requirements of food security and sustainability, the system offers an integrated solution to some of the most controversial issues in the food supply chain. With increased research and development, it can be implemented on a large scale in Bangladesh and the world as a whole, conserving food, making people eat healthier food, and improving sustainable processes in the food industry.

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